Developing Case-Based Reasoning for Structural Design

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CASE-BASED SYSTEMS ENABLE USERS TO RETRIEVE PREVIOUSLY KNOWN DESIGNS FROM MEMORY AND ADAPT THEM TO FIT THE CURRENT DESIGN PROBLEM. THE FOUR CASE-BASED DESIGN SYSTEMS DESCRIBED HERE ILLUSTRATE HOW VARIOUS IMPLEMENTATIONS ACHIEVE DESIGN ASSISTANCE OR DESIGN AUTOMATION OBJECTIVES.

CASE-BASED REASONING is a problem-solving technique that makes analogies between new problems and previously encountered situations (cases) relevant to solving the problem. Using CBR as a design process model involves the subtasks of recalling previously known designs from memory and adapting these design cases or subcases to fit the current design context.

基于案例的推理是一种问题求解技术，它通过在当前问题与先前遇到的、与该问题求解相关的场景（案例）之间建立类比来解决问题。将基于案例的推理作为设计过程模型时，其子任务包括从记忆中检索已知的设计方案，并对这些设计方案或其子方案进行适应性调整，以使其适用于当前的设计情境。

The detailed development of this process model for a particular design domain proceeds in parallel with the development of the case representation, the case memory organization, and the necessary design knowledge. The selection of an information-representation paradigm and the details of its use for a problem-solving domain depend on the intended use of the information, the project information available, and the nature of the domain.

该过程模型在特定设计领域中的详细构建与案例表示及案例存储组织的开发同步进行，以及必要的设计知识。信息表示形式的选择及其在某一问题求解领域中的具体应用细节，取决于该信息的预期用途、可获得的项目信息以及该领域的性质。

CBR could be used to develop and implement a CBR system. Although this sounds circular, if CBR is a viable approach to problem solving, then it can be applied to the development of the reasoning system itself. Toward that end, this article presents four "cases" of case-based building design systems that we developed at the University of Sydney: **CaseCAD**, **CADsyn**, **Win**, and **Demex**. These systems exemplify alternative case memory contents and organizations and provide insight into different potential implementations of the recall and adaptation subprocesses.

CBR可用于开发和实施构建一个基于案例的推理系统。尽管这听起来循环的，如果CBR是一种可行的故障排除方法，在问题求解过程中，它可应用于推理系统本身的开发。为此，本文提出了四个基于案例的建筑设计系统的“案例”我们在悉尼大学开发的工具：CaseCAD、CADsyn 和 Win 以及 Demex。这些系统体现了替代性有效案例的内存内容与组织结构并为不同的潜在可能性提供见解召回与适应子过程的实现。

Implementing case-based structural design

应用基于案例的结构设计

Developing a CBR system starts with the identification of cases that can provide the basis for solving design problems. We began our CBR projects by locating a source of information on structural design projects—a well-known engineering consulting firm that has worked on many medium- to high-rise buildings in Sydney. The firm gave us access to documented project information—primarily design drawings—and the engineers who worked on the projects. Initially, this seemed to be an ideal situation because the engineers were readily available and open to the idea of developing a case base from their projects.

构建基于案例的推理系统首先需要识别能够为设计问题求解提供依据的案例。在开展基于案例的推理项目之初，我们首先确定了一个关于结构设计项目的资料来源——一家在悉尼承建过多栋中高层建筑的知名工程咨询公司。该公司向我们提供了已归档的项目资料，主要包括设计图纸，以及参与这些项目的工程师的相关信息。起初，这一这似乎是一种理想的情况，因为工程师们随时待命，并且乐于接受我们基于其项目构建案例库的设想。

As it turned out, collecting information about the project designs was more difficult than collecting the specifications to the building contractors. In the building industry, documentation primarily communicates construction information rather than the designers' intentions about how the design is to behave. Information about the geometric and material properties of the proposed building is available, but explicit functional or behavioral information is not. Therefore, we had to infer design information based on our theoretical understanding of how structural systems behave.

事实证明，收集项目设计相关资料的难度远高于向建筑承包商索取技术规格书。在建筑行业中，文档主要传递的是施工信息，而非设计师对设计方案功能表现的意图。关于拟建建筑的几何与材料属性等信息尚可获取，但明确的功能或行为特征信息却难以直接获得。因此，我们不得不基于对结构体系行为特性的理论认知来推断设计信息。

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We also found that before considering each project, we needed a template outlining the design description. That is, we had to decide ahead of time how to represent the case in memory and then try to match our expectations with the information actually available for a specific design project. Collecting case information is an iterative process that moves from collecting information to changing what we were looking for. So, although we thought that CBR system development starts with identifying cases, we discovered that first we had to determine what information we expected the cases to include and what the case-based reasoner's goal would be. At the same time, we had to allow for revisions of our expectations.

我们还发现，在考虑之前在每个项目中，我们都需要一个用于概述设计描述的模板。也就是说，我们必须事先确定如何在内存中表示案例，然后将我们的预期与特定设计项目中实际可获得的信息进行比对。案例信息的收集是一个迭代过程，它从信息的初步采集逐步过渡到对所需信息内容的调整。因此，尽管我们认为基于案例的推理系统开发始于案例的识别，但我们发现，首先必须明确案例应包含哪些信息，以及基于案例的推理机的目标是什么。与此同时，我们还必须预留出对自身预期进行修订的空间。

Some commonalities of our four systems reflect our basic approach to CBR. At the most basic level, each system uses attribute-value pairs to represent case memory. The attribute-value pairs are grouped into subsystems and form hierarchies, both in case memory organization and in indexing. The systems group the attribute-value pairs according to the functional, behavioral, and structural roles they play in design reasoning. Although the basic representation paradigm is simple, the organization of cases and case hierarchies builds a more complex representation.

我们的四个系统在某些方面具有共性，这反映了我们在基于案例的推理（CBR）中所采用的基本方法。在最基本的层面上，每个系统均采用属性–值对来表示案例记忆。这些属性–值对被分组为子系统，并在案例记忆的组织结构与索引机制中形成层级关系。各系统依据属性–值对在设计推理过程中所扮演的功能、行为及结构角色对其进行分组。尽管基本的表示范式较为简单，但通过案例及其层级结构的组织，仍可构建出更为复杂的表示体系。

CaseCAD. The CaseCAD project's objective was to propose and implement a design case representation that helps designers find a set of cases relevant to a specified design problem. CaseCAD is a case-browsing tool: it lets users navigate through its memory to look at information that appears relevant to their current interests. It can also retrieve cases autonomously. To pursue the system objectives, we focused on a multimedia case representation to make information readily understood by designers.[1] We also chose flexible memory-indexing techniques that let designers use many different paths to retrieve a case.[2]

CaseCAD。CaseCAD项目的目标是提出并实现一种设计案例表示方法，以帮助设计师找到与特定设计问题相关的一组案例。CaseCAD是一种案例浏览工具：它允许用户在其知识库中导航，查看与其当前关注点相关的各类信息。此外，该系统还能够自主检索案例。为实现上述系统目标，我们重点采用了一种多媒体案例表示方式，以使设计师能够更直观地理解相关信息。同时，我们还选择了灵活的记忆索引技术，使设计师能够通过多种不同的路径来检索案例。2

Case representation. We collected CaseCAD's design case base for a year, in parallel with system development. We discarded much of the collected data because it was too detailed for solving new conceptual design problems. Finally, we reviewed and collected data from 14 design projects. We implemented four of these projects in the final system to demonstrate recall and adaptation. We did not implement the remaining 10 because they were not necessary to demonstrate these subtasks.

我们在系统开发的同时，历时一年收集了Case-CAD的设计案例库。由于所收集的数据过于详尽，难以直接用于解决新的概念设计问题，我们剔除了其中的大部分数据。

最终，我们从14个设计项目中筛选并整理出了可用数据，并在最终系统中实现了其中的4个项目，以验证案例检索与适应功能。剩余的10个项目则未予实现，因为……

它们对于证明这些子任务而言并非必要。

Graphical representations play a vital role in the communication of design information among designers and other building professionals. Graphical media provide a convenient way to handle the complexity of the design description. The human visual system can rapidly capture the wealth of information implicitly and explicitly represented in images, sketches, and drawings. But there are two problems with relying entirely on graphical case representations. First, a graphical representation does not make the design's function or behavior explicit—it addresses only what the design looks like. Second, computers do not reason about complex graphical representations easily. Some case-based design systems, such as Fabel,[3] that use graphical representations for the user interface perform reasoning by converting the images to alternative symbolic representations.

图形化表达在设计师与其他建筑专业人员之间传递设计信息的过程中发挥着至关重要的作用。图形媒介为处理设计描述的复杂性提供了一种便捷的途径。人类视觉系统能够快速地捕捉图像、草图和图纸中隐含及显性的丰富信息。然而，完全依赖图形化的案例表示也存在两个问题：其一，图形化表达并未明确说明设计的功能或行为——它仅描述了设计的外观；其二，计算机无法基于图形进行推理。

能够轻松处理复杂的图形表示。一些基于案例的设计系统，例如Fabel，其用户界面采用图形表示，并通过将这些图像转换为替代的符号表示来进行推理。

> *THE ATTRIBUTE CATEGORIZATION PROVIDES AN EFFECTIVE WAY OF REASONING ABOUT THE VARIOUS DESIGN VARIABLES, BECAUSE EACH CLASS OF VARIABLES HAS A DIFFERENT SEMANTICS ASSOCIATED WITH IT.*

CaseCAD uses an object-oriented, multimedia case representation. Each object includes a reference to CAD drawings and images, groups of attribute-value pairs, and text-based case descriptions. Cases are decomposed into subcases for easy partial recall and design part reuse. The system uses the attribute-value data for case organization and recall, and the graphics and text descriptions help the designer understand the case information. The attribute-value data also helps designers by summarizing important information in discrete packages and explicitly interpreting the graphical representations.

CaseCAD采用面向对象的多媒体案例表示方法。每个对象包含对CAD图纸和图像的引用、若干属性-值对以及基于文本的案例描述。为便于部分信息的检索与设计模块的重用，案例被分解为子案例。系统利用属性-值数据进行案例的组织与检索，而图形与文本描述则有助于设计者理解案例信息。此外，属性-值数据通过将重要信息以离散的结构化形式加以归纳，并对图形表达进行明确的解释，从而进一步辅助设计工作。

CaseCAD's object-based representation follows Gero's design prototype framework,[4] which classifies attributes by function, behavior, and structure. CaseCAD adds a fourth category, *relation attributes*, to facilitate hierarchies and other relational information. Function attributes represent the design's intended purpose—for instance, `support-building-type`, `grid-geometry`, and `resist-wind-load`. Behavior attributes represent the design's response to its environment (its performance); examples are `lateral-displacement`, `total-useable-area`, and `construction-cost`. Structure attributes are the design's physical features, such as `number-of-floors`, `width`, and `height`. Finally, relation attributes represent relationships among design cases, or between design cases and general concepts.

CaseCAD的基于对象的表示方法遵循Gero的设计原型框架；该框架将属性按功能、行为和结构进行分类。CaseCAD还增加了一个第四类，*relation attributes*，以促进实现层次结构及其他关系信息。功能属性表征设计的预期用途，例如支撑构件类型——网格几何，以及抗风性1载荷。行为属性表征设计对其环境的响应（即其性能），示例包括位移、总可用性等。面积及建设成本。结构固有属性是设计的物理特征，例如楼层数等，宽度和高度。最后，关系属性表示设计案例之间的关系，或设计案例与一般概念之间的关系。

The attribute categorization provides an effective way of reasoning about the various design variables because each class of variables has a different semantics associated with it. Making the distinctions explicit allows the system to deal with each type differently. Also, meaningful categorization of the vast number of variables that typically describe a building's structural design reduces the representation's complexity.

属性分类为各类设计变量的推理提供了一种有效途径，因为每一类变量都具有与其相关的不同语义。明确区分这些类别使系统能够针对每种类型采取不同的处理方式。此外，对通常用于描述建筑结构设计的大量变量进行有意义的分类，也有助于降低表示的复杂性。

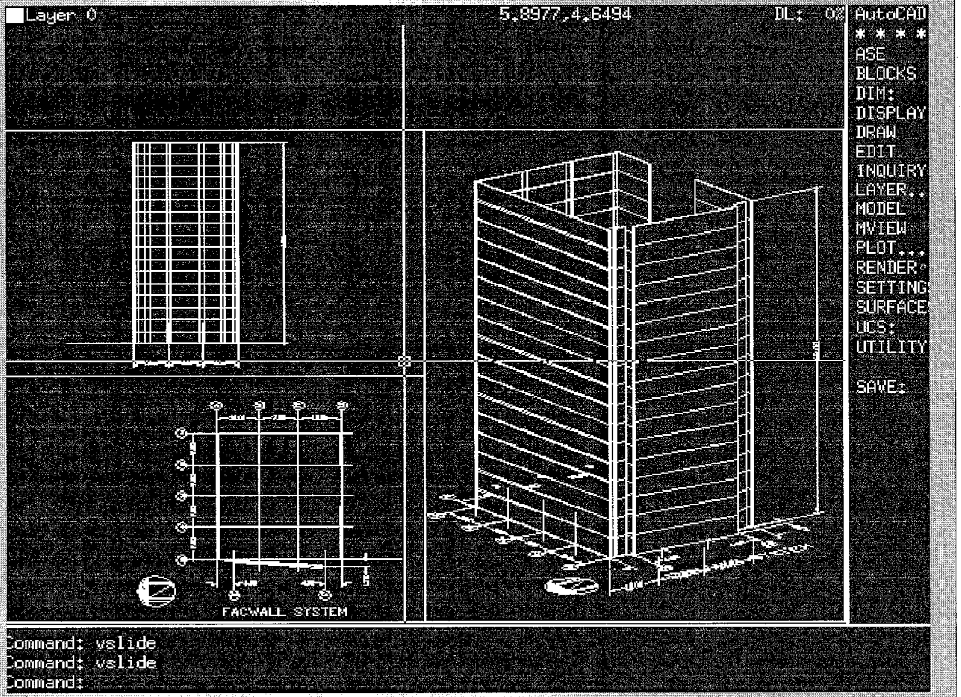
We implemented the case representation in CaseCAD using Framekit, a frame-based representation language derived from Common Lisp. CaseCAD's graphical case representation uses AutoCAD, a general-purpose modeling tool for 3D design visualization, and XFIG, a 2D drawing tool. In CaseCAD, AutoCAD communicates a building design's structural aspects to the user, and XFIG illustrates behavioral information about a case. Figure 1a shows a CaseCAD AutoCAD drawing of a building's structure. Figure 1b shows an XFIG drawing of the same structure's behavior. Figure 1c shows an object-oriented case representation.

我们在CaseCAD中采用Framekit实现了案例表示，Framekit是一种基于框架的表示语言，源自Common Lisp。CaseCAD的图形化案例表示同时使用了AutoCAD——一种用于三维设计可视化的通用建模工具，以及XFIG——一种二维绘图工具。在CaseCAD中，AutoCAD向用户传达建筑设计的结构信息，而XFIG则用于展示案例的行为信息。图1a展示了CaseCAD中利用AutoCAD绘制的建筑结构图；图1b展示了同一结构的行为示意图；图1c则展示了面向对象的案例表示形式。

In addition to knowledge about specific designs in the form of cases, CaseCAD's memory contains general knowledge about classes of designs, in the form of models. For instance, one model represents CaseCAD's knowledge about medium-rise office buildings. CaseCAD models use object-based representation only, not graphical representation. The object-based model representation reflects the case representation—that is, it consists of sets of attributes in three categories. However, a model's attributes can take values from a finite list or range of values, rather than being associated with one specific value. For instance, the attribute `number-of-floors` has a value of 30 in the `Inn-on-the-Park` case representation, but the

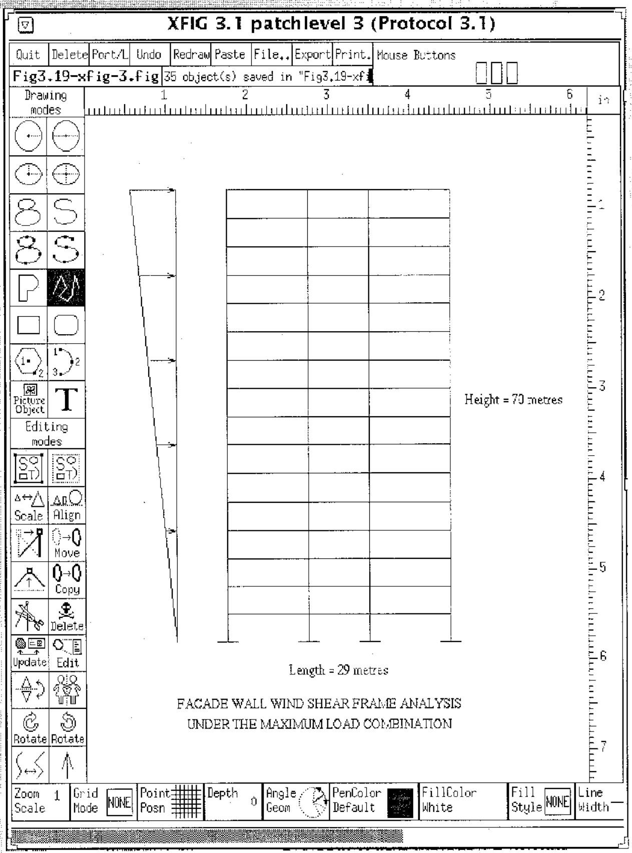
除了以案例形式存储的特定设计知识外，CaseCAD 的记忆库还以模型的形式存储关于各类设计的通用知识。例如，其中一个模型代表了 CaseCAD 对中层办公楼的认知。CaseCAD 的模型仅采用基于对象的表示方式，而不使用图形化表示。这种基于对象的模型表示与案例表示相对应，即由三类属性的集合构成。然而，模型中的属性可以取有限列表或取值范围内的任意值，而无需与某一特定值绑定。例如，在“公园旁酒店”这一案例表示中，属性“楼层数量”或” 的取值为30，但该

Figure 1. Representations used by CaseCAD: (a) AutoCAD drawing of a building's structure; (b) XFIG drawing of a building's behavior; (c) excerpt of an object-oriented case representation.


Layer 0
5.8977,4.6494
DL: 0%
AutoCAD
* * * *
ASE
BLOCKS
DIM:
DISPLAY
DRAW
EDIT
INQUIRY
LAYER...
MODEL
MVIEW
PLOT...
RENDER
SETTING
SURFACE
UCS:
UTILITY
SAVE:
FACWALL SYSTEM

```
Command: vslide
Command: vslide
Command:
```

(a)


XFIG 3.1 patchlevel 3 (Protocol 3.1)
Quit
Delete
Port/L
Undo
Redraw
Paste
File..
Export
Print..
Mouse Buttons
Drawing modes
Picture Object
T
Editing modes
Scale
Align
Move
Copy
Delete
Update
Edit
Rotate
Rotate
Zoom Scale 1
Grid Mode NONE
Point Posn
Depth 0
Angle Geom
PenColor Default
FillColor White
Fill Style NONE
Line Width
1
2
3
4
5
6
in
Height = 70 metres
Length = 29 metres
FACADE WALL WIND SHEAR FRAME ANALYSIS
UNDER THE MAXIMUM LOAD COMBINATION

```
Fig3.19-xfig-3.fig 35 object(s) saved in "Fig3.19-xfi
```

(b)

```
SUBCASE1001
DESCRIPTION: The floor system is a com-
bination of standard Rescrete precast
Formplank panels supported by custom-
made precast beam shells and by facade
wall panels. The ends of the Formplank
panels rested on either corbels on the
facade wall units or the upturned edges
of the precast beam shells. These pre-
cast elements were used to produce a
one-way slab, one-way beam system with
structural detailing providing continu-
ity berween beams, precast columns, and
facade panels.
RELATION
   part-of: 370Pitt
   instance-of: floor-system
FUNCTION
   transmit-dead-load: 3 kPa
   transmit-live-load: 3 kPa
BEHAVIOR
   fire-resistance-level: 2 hrs
   deflection: 18 mm
   shear-stress: 0.98 Mpa
   xfig-file: "370Pitt/floor-system.fig"
STRUCTURE
   material: precast-concrete
   start-level: 1
   end-level: 17
   span-type: one-way
   maximum-span: 11 m
   typical-span: 9 m
   floor-depth: 250 mm
   span-to-depth-ratio: 44
   maximum-aspect-ratio: 4.2
   acad-file: "370Pitt/floor-system.sld"
```

(c)

```
FLOOR-SYSTEM
RELATION
 part-of: building
 instances: subcase1001, ...
FUNCTION
 transmit-dead-load: 2.9-3.1 kPa
 transmit-live-load: 2.9-3.1 kPa
BEHAVIOR
 deflection: 17-19 mm
 fire-resistance-level: 2
STRUCTURE
 material: precast-concrete, reinforced-concrete
 span-type: one-way
```

Figure 2. Partial CaseCAD design model.

图2. CaseCAD设计模型的局部视图。

same attribute has a range of 14 to 30 in the medium-rise office building model representation. In the sample subcase in Figure 1c, the relation attribute `instance-of` indicates the model of the general type of structure the subcase belongs to—specifically, `floor-system`. Every case in CaseCAD is an instance of a model known by the system. Figure 2 shows a partial design model.

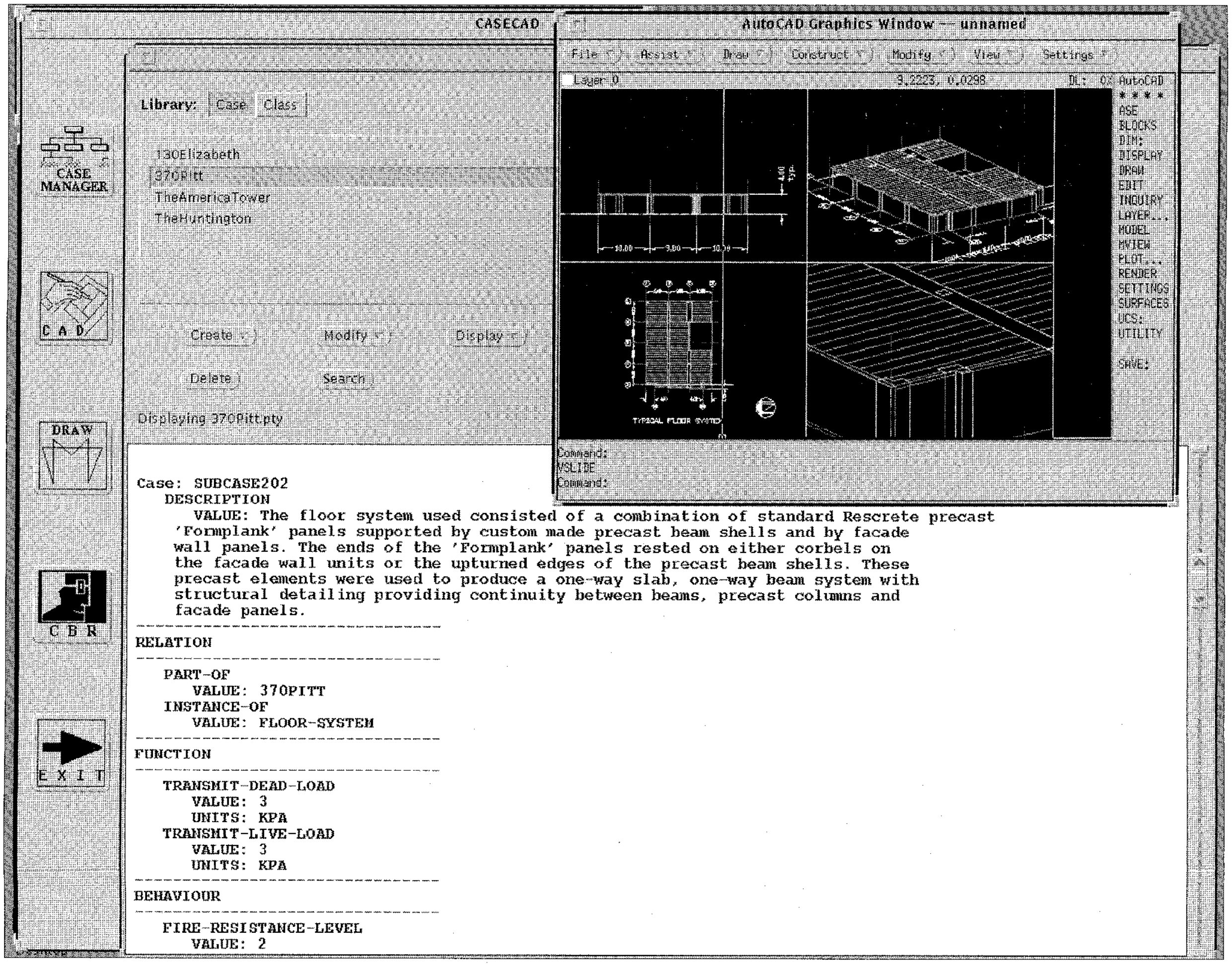
同一属性在中层办公楼模型的表示中取值范围为14至30。在图1C所示的示例子案例中，关系属性实例-o 表明该子案例所属结构类型的一般模型，具体为f l o o r -体系。CaseCAD中的每一个案例都是一个系统已知的模型实例。图2展示了一个部分设计模型。

Browsing and retrieving. CaseCAD's browsing mode enables the user to see the graphical information about a case, using AutoCAD or XFIG, or the object-based representation of the same case. A graphical interface provides menus listing names of cases known to the system, allowing the user to move from one case to another. Figure 3 shows the browser displaying both graphical and text descriptions of the case.

CaseCAD的浏览模式使用户能够以AutoCAD或XFIG格式查看案件的图形信息，或以基于对象的表示形式进行查看同一案例。图形用户界面提供菜单，列出系统已知的案例名称，使用户能够在不同案例之间进行切换。图3展示了浏览器同时显示该案例的图形描述和文本描述。

In the retrieving mode, the user gives the system a design problem specification, which CaseCAD uses as an index to retrieve relevant cases from case memory. Design problem specifications, like the object-based case descriptions and models, consist of a set of attribute-value pairs representing the functional, behavioral, and structural requirements of new structural designs. The system measures a model's or a case's relevance to a given problem by finding matches between the attributes and values in the problem specification and those in the object-based description of the memory item.

在检索模式下，用户向系统提供一个设计问题规范，CaseCAD以此作为索引，从案例库中检索相关案例。设计问题规范与基于对象的案例描述和模型一样，均由一组属性—值对构成，用于表征新结构设计的功能、行为及结构需求。系统通过比对问题规范中的属性及其取值与存储项基于对象的描述中的相应属性及其取值之间的匹配程度，来衡量某一模型或案例与给定问题的相关性。

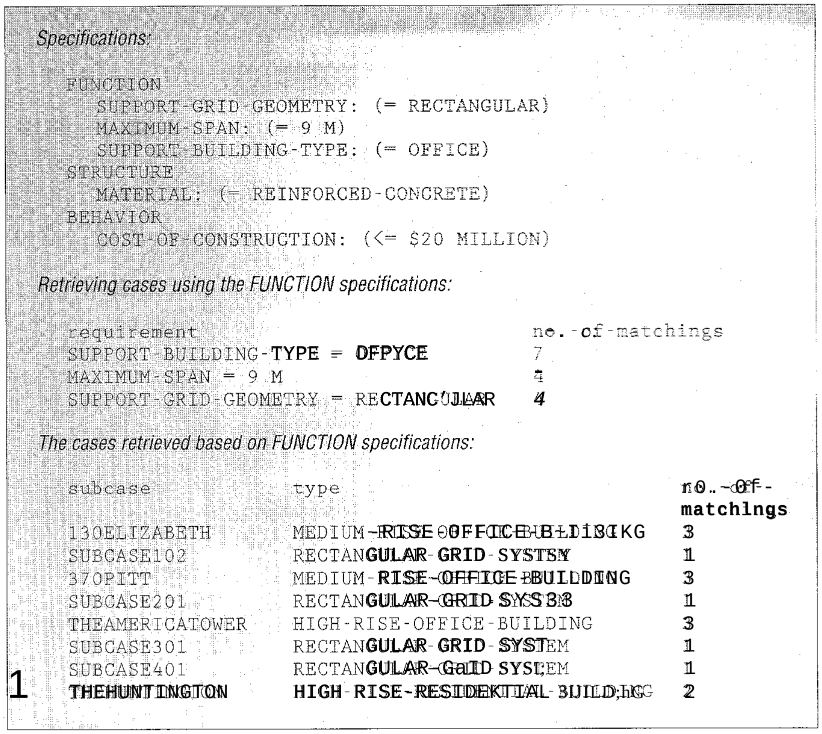

CASECAD
AutoCAD Graphics Window -- unnamed
File
Assist
Draw
Construct
Modify
View
Settings
Layer 0
9.2223, 0.0298
AutoCAD
* * * *
ASE
BLOCKS
DIM:
DISPLAY
DRAW
EDIT
INQUIRY
LAYER...
MODEL
MVIEW
PLOT...
RENDER
SETTINGS
SURFACES
UCS:
UTILITY
SAVE:
TYPICAL FLOOR SYSTEM
Library:
Case
Class
130Elizabeth
370Pitt
TheAmericaTower
TheHuntington
Create
Modify
Display
Delete
Search
CASE MANAGER
C A D
DRAW
C B R
E X I T

```
Command:
VSLIDE
Command:
```

Displaying 370Pitt.pty

```
Case: SUBCASE202
   DESCRIPTION
      VALUE: The floor system used consisted of a combination of standard Rescrete precast
      'Formplank' panels supported by custom made precast beam shells and by facade
      wall panels. The ends of the 'Formplank' panels rested on either corbels on
      the facade wall units or the upturned edges of the precast beam shells. These
      precast elements were used to produce a one-way slab, one-way beam system with
      structural detailing providing continuity between beams, precast columns and
      facade panels.
---------------------------------
RELATION
---------------------------------
   PART-OF
      VALUE: 370PITT
   INSTANCE-OF
      VALUE: FLOOR-SYSTEM
---------------------------------
FUNCTION
---------------------------------
   TRANSMIT-DEAD-LOAD
      VALUE: 3
      UNITS: KPA
   TRANSMIT-LIVE-LOAD
      VALUE: 3
      UNITS: KPA
---------------------------------
BEHAVIOUR
---------------------------------
   FIRE-RESISTANCE-LEVEL
      VALUE: 2
```

Figure 3. Browsing a case in CaseCAD.



```
Specifications:

  FUNCTION
    SUPPORT-GRID-GEOMETRY: (= RECTANGULAR)
    MAXIMUM-SPAN: (= 9 M)
    SUPPORT-BUILDING-TYPE: (= OFFICE)
  STRUCTURE
    MATERIAL: (= REINFORCED-CONCRETE)
  BEHAVIOR
    COST-OF-CONSTRUCTION: (<= $20 MILLION)

Retrieving cases using the FUNCTION specifications:

  requirement                          no.-of-matchings
  SUPPORT-BUILDING-TYPE = OFFICE       7
  MAXIMUM-SPAN = 9 M                   4
  SUPPORT-GRID-GEOMETRY = RECTANGULAR  4

The cases retrieved based on FUNCTION specifications:

  subcase           type                            no.-of-
                                                    matchings
  130ELIZABETH      MEDIUM-RISE-OFFICE-BUILDING     3
  SUBCASE102        RECTANGULAR-GRID-SYSTEM         1
  370PITT           MEDIUM-RISE-OFFICE-BUILDING     3
  SUBCASE201        RECTANGULAR-GRID-SYSTEM         1
  THEAMERICATOWER   HIGH-RISE-OFFICE-BUILDING       3
  SUBCASE301        RECTANGULAR-GRID-SYSTEM         1
  SUBCASE401        RECTANGULAR-GRID-SYSTEM         1
  THEHUNTINGTON     HIGH-RISE-RESIDENTIAL-BUILDING  2
```

Figure 4. Case retrieval in CaseCAD.

图4：在CaseCAD中的案例检索。

The case that matches the largest number of requirements from the problem specification might not be the best case to use for adaptation. The categorization of attributes into function, behavior, and structure provides a way to focus the retrieval. Rather than using explicit weights to assign importance to each specification, CaseCAD lets the user focus on a relevant set of attributes and ignore others while retrieving cases.

与问题规范中最多数量的需求相匹配的案例，未必是最适于进行适应性调整的案例。将属性划分为功能、行为和结构，为检索提供了聚焦的途径。与通过显式权重来赋予各项规范以重要性不同，CaseCAD允许用户在案例检索过程中，重点关注一组相关属性，而忽略其他属性。

Figure 4 shows an example of a user problem specification and the system outputs, detailing cases and subcases that partially match the specification, and the number of requirements they matched. Also shown is the type of structure the case or subcase exemplifies—that is, the model of which the case or subcase is an instance. The retrieval in this example uses only functional attributes as a focus for determining relevance.

图4展示了一个用户问题规格说明及其系统输出的示例，详细列出了与该规格说明部分匹配的案例及子案例，并标明了它们所匹配的需求数量。此外，图中还显示了该案例或子案例的结构类型示例，即，该案例或子案例所属的模型。在本例中，检索仅以功能属性作为确定相关性的依据。

Case adaptation. **CaseCAD** does not perform case adaptation automatically. The designer can modify retrieved cases and store them as new cases in case memory. For the adaptation, the designer uses editing tools to change the object representation, AutoCAD to modify the CAD drawing, and XFIG to modify the images. When the designer adds the case to case memory, CaseCAD automatically indexes the object representation, using the attribute-value pairs defined by the designer.

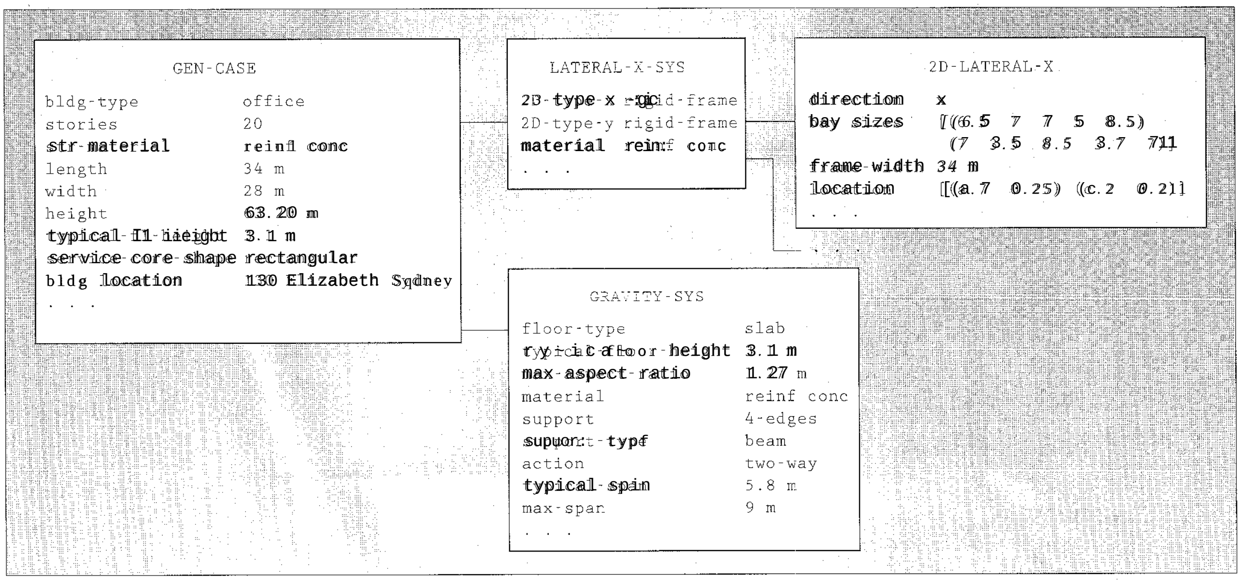
CaseCAD 不会自动执行案例适配。设计者可以对检索到的案例进行修改，并将其作为新案例存入案例库。在适配过程中，设计者利用编辑工具修改对象表示，借助 AutoCAD 修改 CAD 图纸，并使用 XFIG 修改图像。当设计者将该案例添加至案例库时，CaseCAD 会根据设计者定义的属性-值对自动对该对象表示进行索引。

CADsyn. The CADsyn project's objective was to develop a design system that adapts a recalled case automatically to fit a new context.[5] To that end, we developed a case memory representation complementing earlier research on design knowledge representation. The focus on case adaptation was necessary to understand the type and extent of knowledge that would make a CBR system more than a memory-based system. That is, CBR could provide a process model for generating new designs, not just recalling relevant ones.

CADsyn.项目的目标是开发一种设计系统，能够自动将已检索的案例适配至新的情境中。为此，我们构建了一种案例记忆表示方法，以补充先前关于设计知识表示的研究成果。聚焦于案例适配这一环节，有助于厘清使基于案例推理（CBR）系统超越单纯记忆系统的知识类型及其范围。换言之，CBR不仅能够检索相关案例，还能够提供用于生成新设计方案的过程模型。

Case representation. **CADsyn**'s case representation is somewhat similar to CaseCAD's. CADsyn decomposes cases into subcases,

CADsyn 的案例表示方式与 CaseCAD 有一定相似之处。CADsyn 将案例分解为子案例



```
GEN-CASE
bldg-type            office
stories              20
str-material         reinf conc
length               34 m
width                28 m
height               63.20 m
typical-fl-height    3.1 m
service-core-shape   rectangular
bldg location        130 Elizabeth Sydney
. . .
```

```
LATERAL-X-SYS
2D-type-x  rigid-frame
2D-type-y  rigid-frame
material   reinf conc
. . .
```

```
2D-LATERAL-X
direction    x
bay sizes    [(6.5  7  7  5  8.5)
              (7  3.5  8.5  3.7  7)]
frame-width  34 m
location     [(a.7  0.25) (c.2  0.2)]
. . .
```

```
GRAVITY-SYS
floor-type             slab
typical-floor-height   3.1 m
max-aspect-ratio       1.27 m
material               reinf conc
support                4-edges
support-type           beam
action                 two-way
typical-span           5.8 m
max-span               9 m
. . .
```

Figure 5. Part of the case representing 130 Elizabeth Street in CADsyn.

and the primary object-based representation uses attribute-value pairs to describe a case. There are major differences, however. First, because CADsyn focuses on case adaptation rather than browsing and retrieval, it does not use graphical, textual, or other human-amenable representations. Adapting the attribute-value pair representation is possible with knowledge-based techniques, but adapting graphical representations or text descriptions is difficult. Second, CADsyn cases are related to models, but we refer to the models as systems and subsystems. This is due to CADsyn's association with EDEsyn, a representation of design synthesis knowledge using decomposition hierarchies called systems. Finally, CADsyn augments the case and system knowledge representation with constraints to adapt cases using a constraint satisfaction approach.

Most cases in CADsyn's memory are the same as in CaseCAD. CADsyn's memory contains four cases, each with 10 to 14 subcases. The differences between CaseCAD's and CADsyn's case representations lie in the attribute-value pairs and the decomposition into subcases. In CADsyn, we chose attributes and their values according to their utility in implementing the case adaptation subtask and their involvement in constraints. Figure 5 shows an example of a case and a subset of its subcases.

Case retrieval. Design problem specification in CADsyn is the same as in CaseCAD: a set of requirements, in the form of attribute-value pairs, that a design should satisfy to be a valid solution to the problem. The system matches the problem specification with the case descriptions in its memory and retrieves cases and subcases with a partial match. It ranks the retrieved cases according to the number of matching features, favoring the closest match.

CADsyn uses two case-indexing structures: a requirements hierarchy and a subsystem list. The requirements hierarchy uses a conceptual clustering approach.[7] The system traverses the hierarchy according to the problem specification until it finds a list of matching cases. In other words, CADsyn does not search the case memory to match and retrieve cases, but rather searches a separate representation of the requirements hierarchy, which points to relevant cases. The subsystem list associates a subsystem label with a set of subcases that use that label. If CADsyn does not find a relevant case in the requirements hierarchy, it sends the problem specification to the decomposition reasoner, which identifies a relevant set of subsystems. CADsyn then uses these subsystems to retrieve associated subcases and performs adaptation on the subcases rather than on an entire building design.

```
CONSTRAINT-1
   (bldg-type) in (office parking institution)
   (gravity-system/floor-type) in (flat-plate flat-slab)
CONSTRAINT-2
   (bldg-type) in (hotel apartment)
   (gravity-system/floor-type) = slab
CONSTRAINT-3
   (gravity-system/floor-type) in (flat-plate flat-slab)
   (gravity-system/action) = one-way
   (gravity-system/typical-span) < 10 m
CONSTRAINT-4
   (gravity-system/floor-type) in (flat-plate flat-slab slab)
   (gravity-system/typical-span) > 9 m
   (gravity-system/action) = two-way
CONSTRAINT-5
   (gravity-system/support) in (4-edges cols-only)
   (gravity-system/max-aspect-ratio) > 1.5
   (gravity-system/action) = two-way
```

Figure 6. CADsyn's constraint representation.

Design case adaptation. **CADsyn** uses constraint satisfaction to adapt design cases, representing constraints as a set of attribute-value pairs that identify an invalid design. If the constraints represented valid assignments of values to variables, they would only help verify the correctness of proposed designs. Using them to represent invalid value assignments helps detect exactly where problems lie in a proposed design, thus focusing attention on design aspects that need alteration to achieve a goal.

Figure 6 shows examples of constraints in CADsyn. `CONSTRAINT-2` can be read as "If the design uses a `floor-type = slab` and `bldg-type = hotel`, then the design is invalid." The constraints are represented as logical relationships among attribute-value pairs, rather than as numerical relationships.

CADsyn's constraint satisfaction modifies a proposed complete design description, based on a retrieved case or subcase, to satisfy the design specifications. Modifying the retrieved case or subcase to match the design specifications typically introduces constraint violations that must be resolved. For example, if a problem specification for a hotel retrieves an office building case, the system must change the value of the attribute `bldg-type` of the retrieved case to `hotel` as the first step in adapting the case to the user's requirements. Additionally, let's assume that the `floor-type` of the retrieved case has a value of `slab` and that the problem specification did not include any reference to the floor type the hotel should have. The value of the attribute `floor-type` in the partially constructed solution would now violate `CONSTRAINT-2`.

Thus, CADsyn's design case adaptation takes an initially inconsistent assignment of values for design attributes and repairs constraint violations through a search process until it achieves a consistent assignment. Reflecting the proposed design's hierarchical representation as a composition of subcases, the constraint satisfaction adaptation generates a feasible design by modifying invalid subsystems (subcases) in the inconsistent potential design. CADsyn detects invalid subsystems by testing constraints and modifies them using its general knowledge of the valid values of design attributes. Because the constraints are not numerical, the system must validate subsystems incrementally, and determining which invalid subsystem to modify first is a central task.

Figure 7 illustrates the subtasks constituting CADsyn's computational model of design case adaptation as constraint satisfaction. These subtasks iterate until they reach a consistent solution to the user's design problem. The figure also shows the knowledge supporting each task. The constraints provide the knowledge necessary to check constraints, find invalid systems, and select a feasible combination of attribute values for a subsystem design. The feasible combination is the combination that violates the fewest constraints. The system uses decomposition knowledge to find invalid systems, to find valid combinations for new design descriptions, and to propagate the effects of the new subsystem design. It uses heuristic rules to select a subsystem to modify. Heuristic rules are necessary because

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而基于对象的主要表示形式采用属性-值对来描述案例。然而，两者之间存在显著差异。首先，由于CADsyn侧重于案例的适应而非浏览与检索，因此不使用图形、文本或其他便于人类理解的表示方式。利用知识工程技术可以对属性-值对表示进行适配，但对图形表示或文本描述的适配则较为困难。其次，CADsyn中的案例与模型相关联，但我们将其称为系统和子系统，这是由于CADsyn与EDEsyn密切相关——EDEsyn是一种采用称为“系统”的分解层次结构来表示设计综合知识的框架。最后，CADsyn通过引入约束条件来扩展案例与系统的知识表示，从而借助约束满足方法实现案例的适配。

CADsyn的存储中所包含的大多数案例与CaseCAD中的案例相同。CADsyn的存储中共有四个案例，每个案例又细分为10至14个子案例。CaseCAD与CADsyn在案例表示上的差异主要体现在属性-值对以及向子案例的分解方式上。在CADsyn中，我们依据属性及其值在实现案例适配子任务中的效用性以及其在约束条件中的参与程度来选择属性及其取值。图5展示了一个案例及其部分子案例的示例。

设计问题规范在CADsyn中，其形式与CaseCAD相同。一组以属性形式表示的需求。数值对，即一个设计方案为该问题的有效解所应满足的条件。系统将其记忆库中的案例描述与问题规范进行匹配，并检索出部分匹配的案例及子案例。系统根据匹配特征的数量对检索到的案例进行排序，优先选择匹配度最高的方案。

CADsyn采用两种案例索引结构：需求层次结构和子系统列表。需求层次结构采用一种概念聚类方法。系统根据问题规格说明遍历该层次结构，直至找到一组匹配的案例。换言之，CADsyn并非直接在案例库中进行匹配与检索，而是通过一个独立的需求层次结构表示来查找相关案例，该结构指向相应的案例。子系统列表则将一个子系统标签与一组使用该标签的子案例关联起来。如果CADsyn在其中未能找到相关案例在需求层次结构中，它将问题规格传递给分解推理机，后者识别出一组相关的子系统；随后，CADsyn利用这些子系统检索相应的子案例，并在子案例层面而非整个建筑设计层面进行适配。

图6. CADsyn的约束表示。

CADsyn 采用约束满足机制来调整设计实例，将约束表示为一组属性-值对，用于标识无效设计。如果这些约束代表的是变量的合法取值，则仅能用于验证所提设计方案的正确性；而将其用于表示非法取值，则有助于精确定位所提设计方案中的问题所在，从而将注意力集中于那些需要修改以达成目标的设计环节。

图6展示了CADsyn中的约束示例。约束“CONSTRAINT 2”可解读为：若设计采用楼板类型=的板结构，且建筑类型为酒店，则该设计无效。”这些约束以属性-值对之间的逻辑关系而非数值关系来表示。

CADsyn的约束求解机制基于已检索的案例或子案例，对所提出的完整设计描述进行修改，以满足设计规范。在将检索到的案例或子案例调整至符合设计规范的过程中，通常会引入需要予以解决的约束违反。例如，若针对酒店的设计问题检索到了一栋办公楼的案例，则系统必须首先将该案例中属性 bldgtype 的值修改为 hotel，从而使其适应用户的需求。此外，假设该检索案例中 floor-type 的取值为 slab，而问题规该定义并未提及酒店应采用的地板类型。在部分构建的解中，属性 floor-type 的取值现将违反约束-%。

因此，CADsyn 的设计实例适配机制接收一个初始状态下设计属性值不一致的赋值，并通过搜索过程修复约束违反，直至获得一个一致的赋值。鉴于所提出的设计方案以子实例的组合形式进行层次化表示，该约束满足适配机制通过修改不一致潜在设计中的无效子系统（即子实例）来生成一个可行的设计。CADsyn 通过测试约束来识别无效子系统，并利用其关于设计属性有效取值的通用知识对其进行修正。由于这些约束并非数值型约束，系统必须以增量方式对子系统进行验证，而确定首先修改哪个无效子系统成为一项核心任务。

图7展示了CADsyn在设计案例适配过程中作为约束满足问题的计算模型所包含的各个子任务。这些子任务不断迭代，直至为用户的设计问题找到一个一致的解决方案。图中还标明了支撑每一项任务的知识。约束条件提供了用于检验约束、识别无效系统以及为子系统设计选择一组可行属性值组合所需的知识。所谓可行组合，是指违反约束最少的那组属性值组合。该系统利用分解知识来识别无效系统，为新的设计描述寻找有效组合，并传播新子系统设计所带来的影响。此外，它还借助启发式规则来选择待修改的子系统。之所以需要启发式规则，是因为

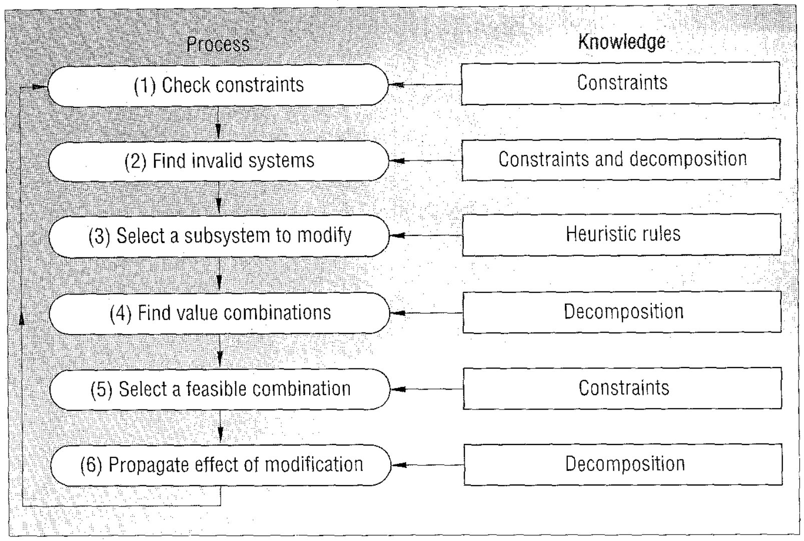

Process
Knowledge
(1) Check constraints
Constraints
(2) Find invalid systems
Constraints and decomposition
(3) Select a subsystem to modify
Heuristic rules
(4) Find value combinations
Decomposition
(5) Select a feasible combination
Constraints
(6) Propagate effect of modification
Decomposition

Figure 7. CADsyn's constraint satisfaction process.

图7. CADsyn的约束满足过程。

the constraints themselves are heuristics, making the design space difficult to analyze.

这些约束本身即为启发式方法，使得设计空间难以进行分析。

The complexity of CADsyn's constraint satisfaction process accommodates adaptation of multiple subcases in a hierarchical design representation in which constraints represent logical relationships among design variables. CADsyn's use of constraints contrasts with other systems. For example, Julia, a case-based reasoner for meal planning, uses constraints to adapt cases, representing the meal as a whole case.[8] Cadre, a case-based reasoner for floor plan design, uses constraints to represent numerical relationships among dimension variables.[9]

CADsyn的约束满足过程具有高度复杂性，能够适应层次化设计表示中的多个子案例，其中约束用于表达设计变量之间的逻辑关系。CADsyn对约束的使用与其他系统形成鲜明对比。例如，Julia作为一种用于膳食规划的基于案例的推理系统，利用约束来调整案例，将整餐视为一个整体案例；而Cadre作为一种用于平面布局设计的基于案例的推理系统，则利用约束来表示尺寸变量之间的数值关系。

Win. The Win project's objective was to develop a case-based design assistant with a structural engineering perspective.[10] The project focused on case memory content for a specific structural design aspect: design and layout of wind-resisting systems in medium- and high-rise buildings. The result is a case memory representation specific to that domain.

Win. 项目的目标是开发一个从结构工程视角出发的基于案例的设计辅助系统。该项目聚焦于某一特定结构设计领域的案例库内容，即中高层建筑抗风体系的设计与布置，其成果是一种针对该领域特有的案例库表示方法。

Case representation. Three of the structural design cases in CaseCAD are represented in the Win system, where each case has approximately four subcases. In Win, cases have more specific information about the engineering performance of lateral load systems than in CaseCAD. Win's case base contains more numerical information about stresses and loads and less high-level symbolic information about the architectural layout. Like CaseCAD, Win's memory contains knowledge in the form of both models and cases, described as attributes and values. Attributes fall into four categories: relational, functional, behavior, and structure. Both models and cases are decomposed into subsystems.

CaseCAD中的三个结构设计案例在Win系统中得以体现，其中每个案例约包含四个子案例。在Win系统中，关于抗侧力体系工程性能的信息比在CaseCAD中更为详尽。Win的案例库包含了更多关于应力与荷载的数值信息，而关于建筑布局的高层次符号化信息则相对较少。与CaseCAD类似，Win的记忆库同样以模型和案例的形式存储知识，被描述为属性与值。属性可分为四类：关系、功能、行为和结构。模型与案例均被分解为子系统。

Models play different roles in CaseCAD and Win. CaseCAD models are decomposed into subsystems that facilitate the acquisition and retrieval of cases. Win models assist case memory organization and problem specification. Specifically, Win models are defined according to two factors that greatly influence the design of a structure's wind-resisting system: the building's dimensions and the position of the building's core system with respect to the rest of its structure. Thus, Win has four models: medium- and high-rise buildings and central- and offset-core structures. These models' primary purpose is to partition case memory according to the intended use.

在CaseCAD和Win中，模型分别发挥不同的作用。CaseCAD中的模型被分解为若干子系统，以促进案例的获取与检索；而Win中的模型则有助于案例库的组织以及问题的精确描述。具体而言，Win中的模型是根据两个对结构抗风体系设计具有重要影响的因素来定义的：建筑的尺度参数以及建筑核心筒的位置。其结构的其余部分而言是独立的。因此，Win具有四种模型：中高层建筑以及中心核心和偏心核心结构。这些模型的主要目的是根据预期用途对案例记忆进行划分。

Figure 8a shows a model corresponding to high-rise structures with an offset core. Figure 8b shows a Win case representation of the Inn on the Park, a 30-story hotel and office building in Sydney. Figure 8c shows how Win uses models to partition case memory.

图8a展示了一个与具有偏心核心的高层建筑类别相对应的模型。图8b展示了“公园酒店”（Inn on the Park）的Win案例表示，这是一栋位于悉尼的30层酒店及办公楼。图8c则说明了Win如何利用模型对案例记忆进行分区。

Indexing and retrieving. Case retrieval in Win is an interactive, iterative process. First, Win prompts the user for a set of problem requirements. For each of several important attributes (`number-of-floors`, `width`, `length`, `floor-to-floor-height`, and so on), Win asks the user to input the value, if known. The resulting set of attribute-value pairs becomes the problem specification, which Win uses as an index to its memory. Win first retrieves models or submodels that match the problem requirements before retrieving cases or subcases that are instances of the models. After model retrieval, Win considers the difference between the user's problem specifications and the extent and range of values of attributes in the models. Win requests additional information, or clarification, before initiating case retrieval.

案例检索在Win中是一个交互式的迭代过程。首先，Win会提示用户输入一组问题需求。对于若干重要属性（如楼层数、宽度、长度、层高等等），Win会提示用户输入其值（如果已知）。由此得到的一组属性-值对即构成问题规范，Win将其用作其知识库的索引。Win首先检索与问题需求相匹配的模型或子模型，然后再检索作为这些模型实例的案例或子案例。在完成模型检索后，Win会评估用户的问题规格与模型中各属性的取值范围及程度之间的差异，并在启动案例检索之前请求用户提供补充信息或进一步说明

This model-based case indexing allows the system to ignore cases and subcases that are completely irrelevant to a problem and efficiently find those that are relevant. Both the model memory and the case memory are flat structures—that is, retrieval proceeds by attempting to match the index with each item in memory successively. The model memory partitions the case memory. The small number of models and cases in Win's memory makes this flat memory organization feasible; more models would make it inefficient. The user always controls the use of Win's model and case bases and the retrieval and selection of cases.

这种基于模型的案例索引机制使系统能够忽略与问题完全无关的案例及子案例，并高效地检索出相关案例。模型记忆库和案例记忆库均采用扁平化结构——即通过依次将索引与记忆库中的每一项进行匹配来实现检索。模型记忆库对案例记忆库进行划分。由于Win的记忆库中模型和案例的数量较少，这种扁平化的记忆组织方式是可行的；若模型数量增多，则会导致效率下降。用户始终掌控Win的模型库和案例库的使用，以及案例的检索与选择过程。

Demex. The Demex (Design by Memory Exploration) project's objective was to consider flexible retrieval and memory exploration.[11] In this project, we developed the strategies of *index elaboration* and *index revision*, which we'll describe in more detail later. With the Demex case memory, a designer can explore and develop an improved design problem specification, rather than develop a design solution. The system reflects real-world design problem solving, in which a designer does not usually start with a well-specified problem and then move toward a solution, but rather the problem requirements evolve as the designer explores alternative solutions.[12]

Demex（“基于记忆探索的设计”）项目的目标是考虑灵活的检索与记忆探索。在该项目中，我们开发了*index elaboration*和*index*两种策略，将在后文予以更详细的阐述。借助Demex案例记忆，设计者可以探索并完善设计问题的规格说明，而非直接着手设计解决方案。该系统反映了现实中的设计问题求解过程：设计者通常并非从一个已明确界定的问题出发进而寻求解决方案，而是随着对备选方案的探索，问题需求逐步演进。”

Case memory representation. Demex uses CaseCAD's design cases and models, partitioning cases and models into subsystems and categorizing attributes as functional, behavioral, structural, and relational. Because of Demex's highly interactive nature, we devised a memory organization that allows efficient and flexible access to cases and subcases.

Demex使用CaseCAD的设计案例与模型，将这些案例与模型划分为子系统，并将属性归类为功能属性、行为属性、结构属性和关系属性。鉴于Demex具有高度交互性的特点，我们设计了一种内存组织方案，以实现对案例及其子案例的高效且灵活的访问。

In Demex, three memories hold conceptually different types of information: an associative memory that relates the function, behavior, and structure categories with design attributes; a model memory; and a case memory. Although the three memories are stored as separate entities in their own data struc-

在Demex系统中，三种记忆分别存储着概念上不同类型的信息：一种关联记忆，用于将功能、行为与结构类别同设计属性相关联；一种模型记忆；以及一种案例记忆。尽管这三种记忆以独立的实体形式分别存储于各自的数据结构中，

tures to increase efficiency and avoid redundancy, they are interrelated. To begin with, we designed all three using the same attribute-value ontology. Additionally, each model in the model memory contains pointers to all cases that are instances of that model, and each case in the case memory contains a pointer to the model of which that case is an instance. Cross-indexing of the semantically different knowledge types allows flexible and efficient traversal among them.

为提高效率并避免冗余，这些要素彼此相互关联。首先，我们采用相同的属性设计了这三者值本体。此外，模型存储中的每个模型都包含指向其所有实例化案例的指针，而案例存储中的每个案例也都包含指向其所属模型的指针。对语义上不同的知识类型进行交叉索引，能够实现它们之间灵活且高效的遍历。

Demex's indexing scheme for model and case memory is a tree of attributes and values (see Figure 9). The system can rapidly retrieve relevant models and cases using the corresponding indexing tree to perform a two-step comparison for each attribute-value pair in the problem specification. The first step traverses the tree along a given attribute arc; the second step traverses it along an arc corresponding to a value for that attribute, retrieving a set of cases with the specified attribute-value pair. If only some categories of attributes are of interest, the retrieval ignores requirements whose attributes belong to an uninteresting category. We designed this three-memory organization and indexing-tree structure especially to support the retrieval strategies we'll describe next.

Demex用于模型和案例记忆的索引方案是一种由属性及其取值构成的树状结构（见图9）。系统可通过相应的索引树，针对问题规约中的每一对属性-取值，执行两步比较，从而快速检索出相关模型与案例。第一步沿给定属性对应的分支遍历该树，第二步则沿该属性取值所对应的分支进行遍历，以检索出具有指定属性-取值组合的一组案例。若仅关注部分属性类别，则案例检索会忽略那些其属性属于非感兴趣类别的约束条件。我们专门设计了这种三重记忆组织及索引树结构，以支持下文将要介绍的检索策略。

Index elaboration and revision. Reconstructing and reconsidering and reformulating a problem's specifications are an important aspect of design problem-solving. Demex guides the user in reformulating structural design requirements by improving the user's understanding of the problem.

重构、重新审视并重新表述问题的规格要求是设计问题求解过程中的一个重要环节。Demex通过提升用户对问题的理解，引导用户对结构设计需求进行重新表述。

Demex uses two problem reformulation strategies to provide a better understanding of a design problem: model-based index elaboration and case-based index revision. These strategies evolved from discoveries we made in the CaseCAD project about the need for flexible case retrieval.[2]

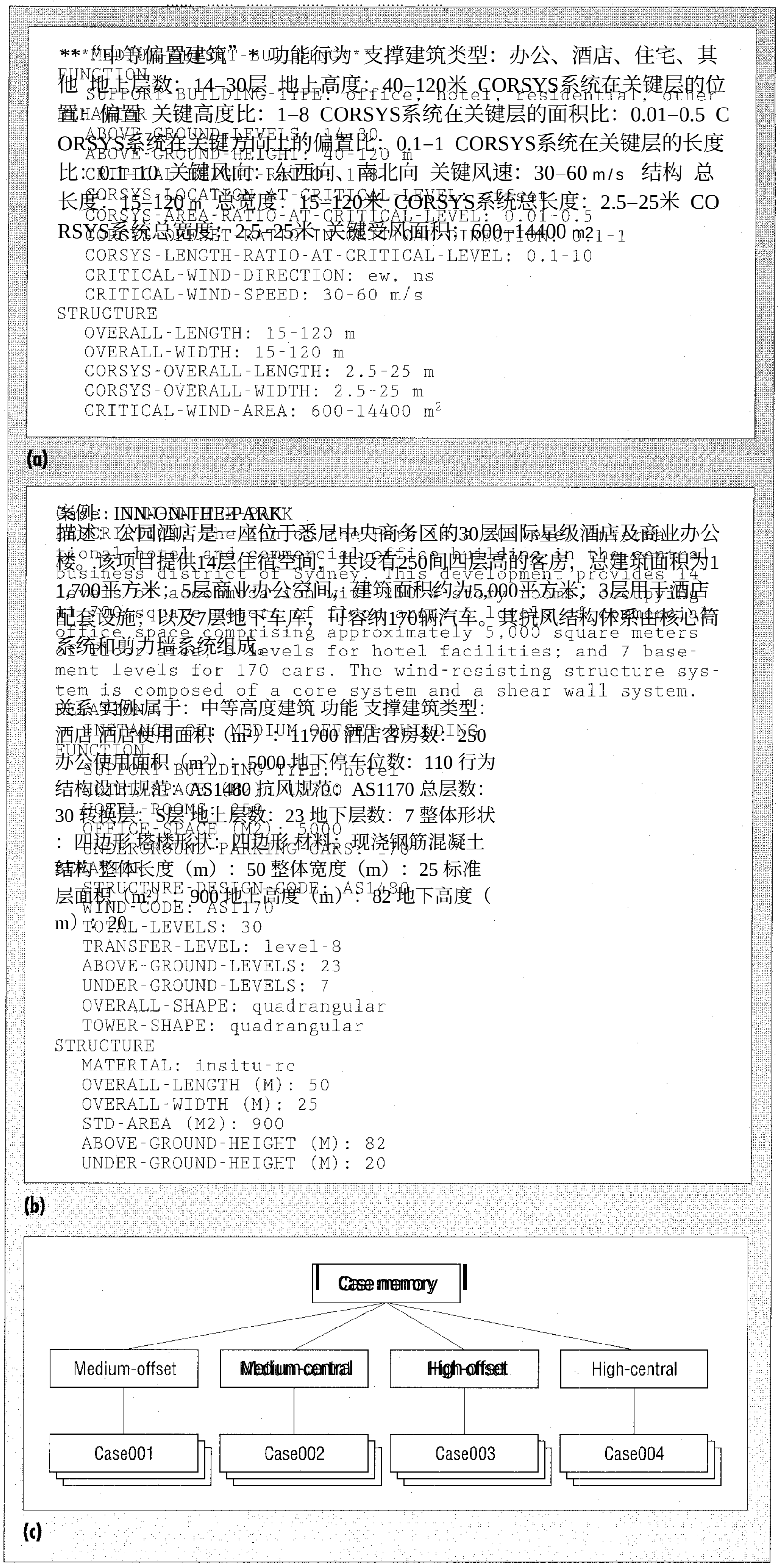
Demex采用两种问题重表述策略，以更深入地理解设计问题：基于模型的指标细化与基于案例的指标修订。这些策略源于我们在CaseCAD项目中所发现的关于灵活案例检索需求的认识。

Making an analogy between the current problem specification and general knowledge in memory, model-based index elaboration retrieves one or more models from memory. The models can provide a better description of the design problem in two ways. First, items retrieved from memory usually contain more information (for example, more design variables to describe a structure) than does the initial problem specification. The designer can expand the set of specifications with this information.

通过将当前的问题描述与记忆中的通用知识进行类比，基于模型的索引细化从记忆中检索出一个或多个模型。这些模型可以从两个方面对设计问题提供更为详尽的描述：首先，从记忆中检索出的条目通常包含比初始问题描述更为丰富的信息（例如，用于描述结构的设计变量数量更多），设计师可利用这些信息扩展问题的规格集。

Second, models provide ranges of acceptable values associated with the design

其次，模型提供了与设计相关的可接受值范围。



```
***MEDIUM-OFFSET-BUILDING***
FUNCTION
  SUPPORT-BUILDING-TYPE: office, hotel, residential, other
BEHAVIOR
  ABOVE-GROUND-LEVELS: 14-30
  ABOVE-GROUND-HEIGHT: 40-120 m
  CRITICAL-HEIGHT-RATIO: 1-8
  CORSYS-LOCATION-AT-CRITICAL-LEVEL: offset
  CORSYS-AREA-RATIO-AT-CRITICAL-LEVEL: 0.01-0.5
  CORSYS-OFFSET-RATIO-IN-CRITICAL-DIRECTION: 0.1-1
  CORSYS-LENGTH-RATIO-AT-CRITICAL-LEVEL: 0.1-10
  CRITICAL-WIND-DIRECTION: ew, ns
  CRITICAL-WIND-SPEED: 30-60 m/s
STRUCTURE
  OVERALL-LENGTH: 15-120 m
  OVERALL-WIDTH: 15-120 m
  CORSYS-OVERALL-LENGTH: 2.5-25 m
  CORSYS-OVERALL-WIDTH: 2.5-25 m
  CRITICAL-WIND-AREA: 600-14400 m²
```

***“中等偏置建筑” * 功能行为 支撑建筑类型：办公、酒店、住宅、其他 地上层数：14–30层 地上高度：40–120米 CORSYS系统在关键层的位置：偏置 关键高度比：1–8 CORSYS系统在关键层的面积比：0.01–0.5 CORSYS系统在关键方向上的偏置比：0.1–1 CORSYS系统在关键层的长度比：0.1–10 关键风向：东西向、南北向 关键风速：30–60 m/s 结构 总长度：15–120 m 总宽度：15–120米 CORSYS系统总长度：2.5–25米 CORSYS系统总宽度：2.5–25米 关键受风面积：600–14400 m2

(a)

```
CASE: INN-ON-THE-PARK
DESCRIPTION: The Park Hotel is a 30-story interna-
tional hotel and commercial office building in the central
business district of Sydney. This development provides 14
levels of accommodation with 250 four-story rooms, occupying
11,700 square meters of floor area; 5 levels of commercial
office space comprising approximately 5,000 square meters
of floor area; 3 levels for hotel facilities; and 7 base-
ment levels for 170 cars. The wind-resisting structure sys-
tem is composed of a core system and a shear wall system.
RELATION
  INSTANCE-OF: MEDIUM-OFFSET-BUILDING
FUNCTION
  SUPPORT-BUILDING-TYPE: hotel
  HOTEL-SPACE (M2): 11700
  HOTEL-ROOMS: 250
  OFFICE-SPACE (M2): 5000
  UNDERGROUND-PARKING-CARS: 170
BEHAVIOR
  STRUCTURE-DESIGN-CODE: AS1480
  WIND-CODE: AS1170
  TOTAL-LEVELS: 30
  TRANSFER-LEVEL: level-8
  ABOVE-GROUND-LEVELS: 23
  UNDER-GROUND-LEVELS: 7
  OVERALL-SHAPE: quadrangular
  TOWER-SHAPE: quadrangular
STRUCTURE
  MATERIAL: insitu-rc
  OVERALL-LENGTH (M): 50
  OVERALL-WIDTH (M): 25
  STD-AREA (M2): 900
  ABOVE-GROUND-HEIGHT (M): 82
  UNDER-GROUND-HEIGHT (M): 20
```

案例：INN-ON-THE-PARK

描述：公园酒店是一座位于悉尼中央商务区的30层国际星级酒店及商业办公楼。该项目提供14层住宿空间，共设有250间四层高的客房，总建筑面积为11,700平方米；5层商业办公空间，建筑面积约为5,000平方米；3层用于酒店配套设施；以及7层地下车库，可容纳170辆汽车。其抗风结构体系由核心筒系统和剪力墙系统组成。

关系 实例属于：中等高度建筑 功能 支撑建筑类型：酒店 酒店使用面积（m²）：11700 酒店客房数：250 办公使用面积（m²）：5000 地下停车位数：110 行为 结构设计规范：AS1480 抗风规范：AS1170 总层数：30 转换层：S层 地上层数：23 地下层数：7 整体形状：四边形 塔楼形状：四边形 材料：现浇钢筋混凝土 结构 整体长度（m）：50 整体宽度（m）：25 标准层面积（m²）：900 地上高度（m）：82 地下高度（m）：20

(b)

(c)

Figure 8. Models and cases in Win: (a) design model representation; (b) case representation; (c) using models to partition case memory.

图8. Win中的模型与案例：（a）设计模型表示；（b）案例表示；（c）利用模型对案例内存进行分区。

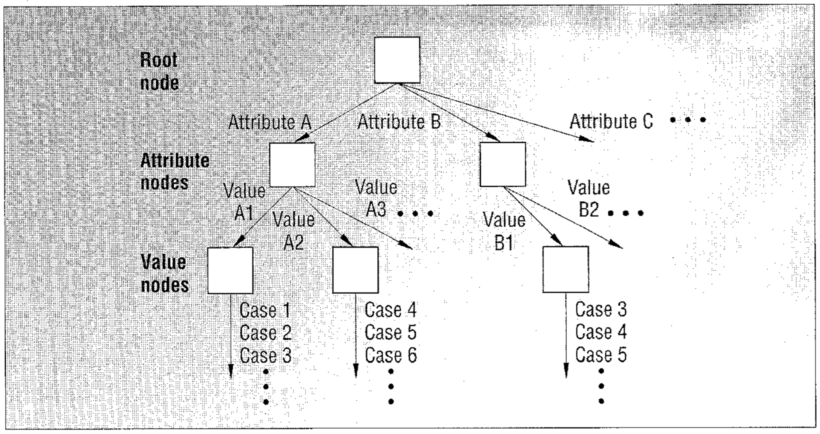


Figure 9. Demex case indexing tree.

图9. Demex案例索引树。

variables that describe the model. The user compares the values in the problem specification and the retrieved model's range of values. Attributes for which the specification value is not in the acceptable range are critical attributes. The user can prune the noncritical parts of a problem specification to focus on the important parts, resulting in a smaller design space.

用于描述模型的变量。用户将问题规范中的取值与所检索模型的取值范围进行比对。对于规范中给定的取值超出可接受范围的属性，即为关键属性。用户可通过剔除非关键部分的问题规范，聚焦于重要部分，从而缩小设计空间。

Figure 10 illustrates model-based index elaboration. The user, interested in designing a medium-height office building, describes the problem to the system (see Figure 10a). Demex uses these problem specifications to retrieve a relevant model. For this example, the best match in the model memory is `medium-rise-office-buildings`, parts of which are shown in Figure 10b.

图10展示了基于模型的索引细化过程。用户希望设计一栋中层办公楼，便向系统描述了该问题（见图10a）。Demex利用这些问题规格从模型库中检索出一个相关模型。在本例中，模型库中的最佳匹配为“medium-rise-office-buildings”，其中的部分如图10b所示。

The retrieved model has more attributes than the user gave in the problem specifications—for example, `location-of-core`, `material`, and `height-to-width-ratio`. If the values of any of these attributes are known, adding them to the problem requirements can improve the focus of case retrieval. Let's assume that Figure 10c shows the user's new problem specifications, with relevant attributes (and their values) added.

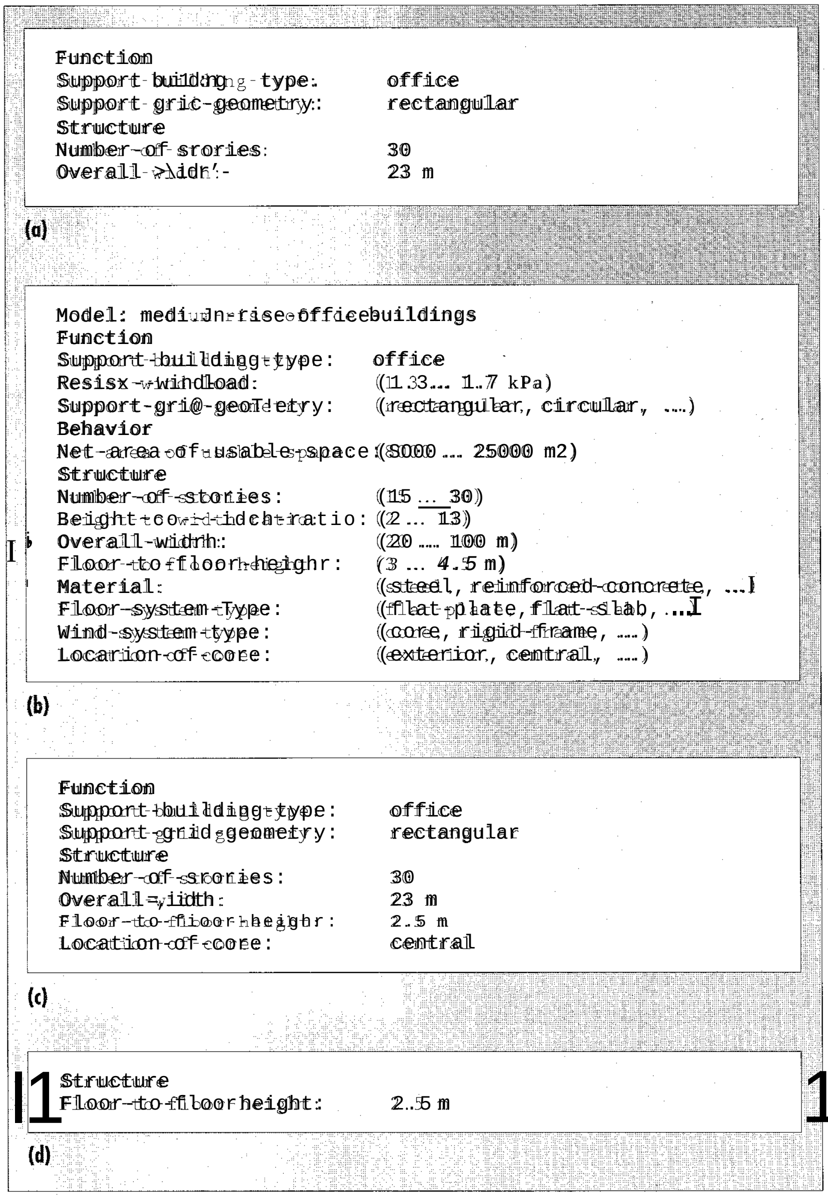
所检索到的模型具有比用户在问题规范中给出的更多的属性，例如，的核心位置，对应，材料，以及高宽比。如果这些属性中的任何值已知，将其纳入问题需求可有助于提高案例检索的针对性。假设图10c展示了用户的新问题规格，并添加了相关属性（及其取值）。

A comparison of the problem specifications, version 2, with the design model shows that a value of 2.5 m for the `floor-to-floor-height` attribute is outside the valid range. This identifies a critical attribute that becomes the new problem specifications (see Figure 10d).

将问题规范第2版与设计模型进行对比表明，地板至的值为2.5m。楼层高度属性超出有效范围。这确定了一个关键属性，并将其作为新的问题规范（见图10d）。

The case-based index revision strategy uses cases to modify the problem specifications. Modifications include adding, removing, and altering some or all of the attribute-value pairs in the problem specifications. Altering an attribute-value association could involve modifying the attribute's name, value, or both.

基于案例的索引修订策略利用案例来修改问题规范。这些修改包括在问题规范中添加、删除以及更改部分或全部属性-值对。更改某一属性-值关联可能涉及修改该属性的名称、值，或同时修改两者。



(a)

```
Function
Support-building-type:      office
Support-grid-geometry:      rectangular
Structure
Number-of-stories:          30
Overall-width:              23 m
```

(b)

```
Model: medium-rise-office-buildings
Function
Support-building-type:      office
Resist-wind-load:           (1.33 ... 1.7 kPa)
Support-grid-geometry:      (rectangular, circular, ....)
Behavior
Net-area-of-usable-space:   (8000 .... 25000 m2)
Structure
Number-of-stories:          (15 ... 30)
Height-to-width-ratio:      (2 ... 13)
Overall-width:              (20 .... 100 m)
Floor-to-floor-height:      (3 ... 4.5 m)
Material:                   (steel, reinforced-concrete, ...)
Floor-system-type:          (flat-plate, flat-slab, ...)
Wind-system-type:           (core, rigid-frame, .... )
Location-of-core:           (exterior, central, .... )
```

(c)

```
Function
Support-building-type:      office
Support-grid-geometry:      rectangular
Structure
Number-of-stories:          30
Overall-width:              23 m
Floor-to-floor-height:      2.5 m
Location-of-core:           central
```

(d)

```
Structure
Floor-to-floor height:      2.5 m
```

Figure 10. Model-based index elaboration in Demex: (a) version 1 of problem specifications; (b) parts of best match retrieved from model memory; (c) version 2 of problem specifications; (d) version 3 of problem specifications.

图10. Demex中的基于模型的索引细化：（a）问题规格说明的第1版；（b）从模型记忆中检索到的最佳匹配部分；（c）问题规格说明的第2版；（d）问题规格说明的第3版。

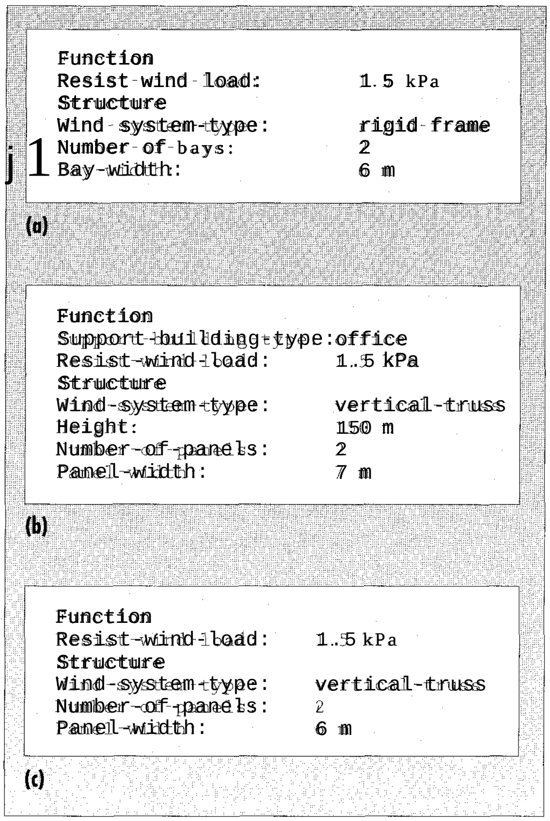
The example in Figure 11, which considers the design of a building's wind-resisting system, illustrates the case-based index revision strategy for problem reformulation. Demex retrieves the case shown in Figure 11b. To achieve wind resistance in structures, the retrieved case uses vertical trusses, not the rigid frames of the original specifications. So, the designer can modify the specifications (see Figure 11c).

图11中的示例以建筑抗风体系的设计为例，说明了用于问题重述的基于案例的索引修订策略。Demex检索出了图1b所示的案例。为实现结构的抗风性能，该检索出的案例采用了竖向桁架，而非原始规范中的刚性框架。因此，设计人员可以对规范进行修改（参见图1c）。

In version 2, the attributes `number-`

在版本2中，属性数量为。

of-bays and bay-width of the specification of the rigid-frame become number-of-panels and panel-width. In one step, some parts of the description have changed, either in their values or their attributes, while others have remained intact. In this example, no parts of the original specification were dropped (although an alternative way of viewing the altering of an attribute-value association is that it was removed and replaced by an analogous association at the same time).



(a)

```
Function
Resist-wind-load:          1.5 kPa
Structure
Wind-system-type:          rigid-frame
Number-of-bays:            2
Bay-width:                 6 m
```

(b)

```
Function
Support-building-type:office
Resist-wind-load:        1.5 kPa
Structure
Wind-system-type:        vertical-truss
Height:                  150 m
Number-of-panels:        2
Panel-width:             7 m
```

(c)

```
Function
Resist-wind-load:        1.5 kPa
Structure
Wind-system-type:        vertical-truss
Number-of-panels:        2
Panel-width:             6 m
```

Figure 11. Case-based index revision in Demex: (a) initial problem specification; (b) retrieved case; (c) second problem specification.

System options

Our four projects and the experiences we gained from them can themselves serve as case studies on the development of case-based design tools. Although we cannot say that one approach is better than another, their differences point to options in developing CBR design systems.

Representation in case-based design. On the surface, all four systems described here use a similar representation paradigm; however, they have important differences. The similarities are the use of attribute-value pairs for representing a case or subcase and the use of models to support CBR. The attribute-value pair description allows indexing and adaptation of specific cases by means of relatively simple pattern matching based on string comparisons of attributes and values. In addition to the attribute-value pairs, the systems (except CADsyn) document cases with drawings, images, and text descriptions. One way of viewing the attribute-value pairs is to consider them the index to the more comprehensive case description.

The systems do not describe cases in terms of lessons learned but in a formalism relevant to design, such as intended function, actual behavior, and structural attributes. The selection of attributes in a case depends on the other knowledge necessary to support CBR. The attributes in a case description must match those in the model descriptions for the system to recognize that the knowledge in the models is relevant to the case. In CADsyn, the attributes in the case representation also must be the same as those included in the constraints used to identify constraint violations.

The differences among the systems include their case memory organization and the way they use indexes for case retrieval. In CaseCAD and CADsyn, memory is organized as lists of cases and models. All attribute-value pairs are valid indexes, and case retrieval consists of matching all or part of the problem specifications with each case and subcase. This approach is appropriate when case memory contains a limited number of cases and inappropriate when case memory becomes very large.

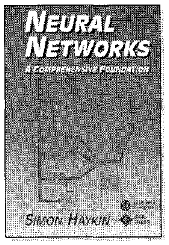
In Win, the models partition case memory so the retrieval process first matches the problem specifications with each model. Assuming there are fewer models than cases, this scheme is more efficient than comparing each case. This approach allows only a comparison of the attribute-value pairs in the models; it is valid when the models adequately describe the functional decomposition of relevant cases and there are few models. The use of models to partition case memory limits retrieval to cases that are similar; it does not allow dissimilar but interesting cases to be retrieved.

Demex's tree structure, with attributes and values as branches, is the most efficient representation of a large case memory, because the number of cases does not affect the time needed to find partial matches.

Each system uses models differently. In CaseCAD, the models provide the template for identifying case information during the acquisition of new cases. CADsyn uses models to perform constraint satisfaction. Each model provides the domain space for the attributes in the case description. These attribute domains are the alternative values for fixing violated constraints. In Win, the models partition case memory and provide a heuristic grouping of indexes for problem elaboration and case retrieval. In Demex, the models provide the space from which index elaboration either refines and adds detail to the problem description or identifies critical requirements. Although the specific roles of models differ in each system, the general role of models is to support the reasoning aspects of case-based design.

Reasoning in case-based design. The purpose of reasoning in case-based design can range from design assistance to design automation. CaseCAD, Win, and Demex focus on design assistance through a memory-based system for the recall of similar designs. The reasoning in these systems relies on the user to guide retrieval, modification of problem specifications, and selection of a case to be pursued further. Recording problem specifications and comparing them to cases and models in memory, the reasoner's main task is memory retrieval.

CADsyn's reasoning aims at design synthesis, which is closer to the design automation end of the spectrum. CADsyn's constraint satisfaction approach directly contrasts with CaseCAD's adaptation approach, in which the user changes the case, unsupported by the case-based reasoner. CADsyn is not a fully automated design system, because the user must select the appropriate case or subcase to be adapted and can override the system's selection of the



Neural Networks: A Comprehensive Foundation

神经网络：全面的基础

Simon Haykin

西蒙·海金 著

Here is the most comprehensive treatment available of neural networks from an engineering perspective. This thorough, well-organized, and completely up-to-date book examines all the important aspects of this emerging technology. The text features comprehensive chapter introductions, numerous computer experiments, network problems and solutions, photographs and illustrations, and a bibliography and thorough glossary that will reinforce the key concepts presented in this book.

本书从工程学视角对神经网络进行了迄今为止最为全面的阐述。这本内容详实、条理清晰且完全与时俱进的著作，系统探讨了这一新兴技术的所有重要方面。书中各章均配有全面的导论，并包含大量计算机实验、网络问题与解决方案、照片与插图，以及一份参考文献和一份详尽的术语表，有助于巩固书中所阐述的核心概念。

Contents: Introduction • Learning Process • Correlation Matrix Memory • The Perceptron • Least-Mean-Square Algorithm • Multilayer Perceptrons • Radial-Basis Function Networks • Recurrent Networks Rooted in Statistical Physics • Self-Organizing Systems: Hebbian, Competitive Learning, and Maximum Information Preservation • Modular Networks • Temporal Processing • Neurodynamics • VLSI Implementations of Neural Networks • Appendixes • Bibliography and References • Index

内容：引言 学习过程 相关性矩阵记忆 感知机 最小均方算法 多层感知机 径向基函数网络 循环网络 根植于统计物理 自组织系统 赫布学习、竞争学习与最大信息保持 模块化网络 时序处理 神经动力学 神经网络的超大规模集成电路实现 附录 参考文献与引用 索引

720 pages. 1994. Hardcover.
ISBN 0-02-352761-7.
Catalog # BP05687
$55.95 Members / $69.95 List

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…reasoning process in design case adaptation, when made explicit, provides alternative design solutions for the designer.

待修改的系统无效。当CADSyn的设计案例适配推理过程被明确化时，它能够为设计者提供备选的设计方案。

THE FOUR SYSTEMS DESCRIBED here do not represent an exhaustive list of options for implementing case-based design. Although we used similar methods to represent design information and the same design cases, a variety of issues led us to develop different systems. We concentrated on structural design representations so that we could address problem formulation and solution generation. Using CBR for problem formulation led us to consider the role of cases in defining the scope of the problem. Using CBR for generating solutions led us to consider what changes and what stays the same in case adaptation, as well as the knowledge needed to make these decisions and methods of automating the process. The practicality of CBR systems for use by professionals depends on further development in design technology such as multimedia database management support, and further developments in design knowledge representation.

本文所描述的四种系统并未穷尽基于案例的设计实现方案。尽管我们采用了相似的方法来表示设计信息并使用了相同的设计案例，但多种因素促使我们开发出了不同的系统。我们着重关注结构化的设计表示，以便处理问题的表述与求解过程的生成。将基于案例的推理用于问题表述，使我们得以思考案例在界定问题范围中的作用；而将其用于求解方案的生成，则使我们深入探讨在案例适配过程中哪些要素需要变更、哪些要素保持不变，以及做出这些决策所需的知识和实现这一过程的自动化方法。基于案例的推理系统在专业领域的实用性，仍有赖于相关技术的进一步发展，例如多媒体数据库管理的支持，以及设计知识表示方法的持续完善。

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